# The quadratic formula

**1** Considering the quadratic formula, find the values of $a$, $b$ and $c$ in the following quadratic equations:

**a** $x^2 + 7x + 10 = 0$

**b** $x^2 - 3x - 4 = 0$

**c** $2x^2 + 9x = 0$

**d** $4x^2 + 3x = 5$

**e** $8x^2 - 3x = 0$

**f** $-8x^2 + 3x = 0$

**g** $-2x^2 + 9x + 5 = 0$

**h** $3x^2 - 8x + 2 = 9x - 7$

**2** Solve the following equations using the quadratic formula:

**a** $x^2 + 5x + 6 = 0$

**b** $x^2 - 5x + 6 = 0$

**c** $x^2 - 5x + \frac{9}{4} = 0$

**d** $x^2 - 8x + \frac{55}{4} = 0$

**e** $2x^2 + 6x - 8 = 0$

**f** $4x^2 - 10x + 4 = 0$

**g** $4x^2 + 7x + 3 = 0$

**h** $4x^2 - 17x - 15 = 0$

**i** $-6 + 7x + 5x^2 = 0$

**j** $-6 - 13x + 5x^2 = 0$

**k** $-20 - 11x + 3x^2 = 0$

**l** $-20 + 21x + 5x^2 = 0$

**3** Solve the following equations, leaving your answer in surd form:

**a** $x^2 - 7x + 9 = 0$

**b** $x^2 - 5x - 2 = 0$

**c** $-2x^2 - 15x - 4 = 0$

**d** $3x^2 + 9x - 4 = 0$

**e** $5x^2 - 15x + 2 = 0$

**f** $-5x^2 - 15x + 3 = 0$

**g** $5x = (x - 5)(3x + 3)$

**h** $\frac{3x + 1}{3x - 1} - \frac{3x - 1}{3x + 1} = 5$

**4** Consider the equation $2x^2 = 14$.

**a** Solve for $x$ using the quadratic equation. Give your answer in surd form.

**b** Give $x$ as decimal number correct to three decimal places.

**5** Solve the following equations, rounding your answers to three decimal places:

**a** $x^2 + 3x - 6 = 0$

**b** $x^2 + 7x - 3 = 0$

**c** $4x^2 + 7x + 2 = 0$

**d** $5x^2 + 9x + 2 = 0$

**e** $2x(x - 4) = 3x + 1$

**f** $3x(x + 4) = -3x + 4$

**g** $7.1x^2 + 5.3x - 1.5 = 0$

**h** $1.8x^2 + 5.2x - 2.3 = 0$

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**6** Solve the following expression for $m$:

$$10 - 6m + 2m^2 = m^2 + 8m + 9$$

**7** For each of the given solutions to a quadratic equation:

**i** Find the values of $a$, $b$ and $c$.

**ii** Write down the quadratic equation that has these solutions.

**a** $x = \dfrac{-5 \pm \sqrt{5^2 - 4 \times (-7) \times 10}}{2 \times (-7)}$

**b** $x = \dfrac{-22 \pm \sqrt{484 - 108}}{18}$

## Applications

**8** An object is launched from a height of $80$ ft with an initial velocity of $107$ ft/s.

After $x$ seconds, its height, $h$, is given by:

$$h = -16x^2 + 107x + 80$$

Find the number of seconds, $x$, after which the object is $30$ ft above the ground. Round your answer to one decimal place.

**9** An object is launched from a height of $8$ m with an initial velocity of $k$ ms$^{-1}$.

After $t$ seconds, its height, $h(t)$, is given by:

$$h(t) = -2t^2 + kt + 8$$

The object was in flight for $4$ seconds before reaching the ground. What was its initial velocity $k$ ?

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